

STAT 516: Some discrete distributions

Poisson distribution

Prof. Michael Levine

February 19, 2024

Some heuristics

- ▶ No particular algorithm that introduces it in a finite number of steps
- ▶ Often, it is a limiting distribution of numerous integer-valued random variables when a sequence of Bernoulli trials makes it harder to obtain a success
- ▶ In other words, a number of times a rare event happens if we observe for a long enough time often has the Poisson distribution

Definition

- ▶ $X \sim \text{Poi}(\lambda)$ if

$$p(x) = P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

for $x = 0, 1, 2, \dots$

- ▶ Easy to verify that $p(x)$ is a pmf by using a Taylor expansion of e^λ

Mean and Variance of the Poisson

- ▶ Let $X \sim Poi(\lambda)$
- ▶ Then,
 1. $E(X) = Var X = \lambda$ - the mean is equal to the variance
 2. $E(X - \lambda)^3 = \lambda$
- ▶ The skewness and curtosis of X are $\beta = \frac{1}{\sqrt{\lambda}}$ and $\gamma = \frac{1}{\lambda}$

Example

- ▶ April receives on average three phone calls per day at her home. What is the percentage of days on which she receives no phone calls?
- ▶ Clearly, if Poisson model is true, $P(X = 0) = e^{-3} = 0.0498$
- ▶ What is the percentage of days she receives more than five phone calls?
- ▶ Again, under the Poisson assumption,
$$P(X > 5) = 1 - P(X \leq 5) = 1 - \sum_{x=0}^5 e^{-3} 3^x / x! = 0.0839$$
- ▶ If using R, use `dpois(0,3)` in the first case and `1-ppois(5,3)` in the second
- ▶ Keep in mind that the Poisson model is not the only one possible!

Example

- ▶ Lengths of an electronic tape contain, on average, 1 defect per 100 ft. What is the probability that the 50 ft long tape will be defect free?
- ▶ Assuming the homogeneous Poisson process, we model it as $X \sim \text{Poi}(0.5)$
- ▶ Thus, $P(X = 0) = e^{-.5} = .6065$

Example

- ▶ Between the months of May and October, one observes a shooting star about once per 20 min
- ▶ You sit on a patio for one hour each evening; how many days would it be before you see ten or more shooting stars on the same day?
- ▶ Let p be the probability of seeing ten or more shooting stars on the same day
- ▶ Let us model the number of shooting stars seen per day s $N \sim Poi(\lambda)$ with $\lambda = 3$

$$p = P(N \geq 10) = \sum_{x=10}^{\infty} \frac{e^{-3} 3^x}{x!} = 0.0011$$

- ▶ If X is the number of days you have to watch the sky until you see ten or more, then $X \sim Geo(p)$ and $E(X) = \frac{1}{p} = 909.1$ - about 30 months

Mode of the Poisson distribution

- ▶ The Poisson distribution satisfies

$$P(X = k) \leq P(X = \lfloor \lambda \rfloor)$$

for any $k \geq 0$

- ▶ Corollary:
 - ▶ If λ is an integer, both λ and $\lambda - 1$ are the modes
 - ▶ If λ is not an integer, $\lfloor \lambda \rfloor$ is the unique mode. for any $k \geq 0$

Poisson Approximation to Binomial

- ▶ Let $X_n \sim \text{Bin}(n, p_n)$ for $n \geq 1$. Suppose $np_n \rightarrow \lambda$ for $0 < \lambda < \infty$ as $n \rightarrow \infty$. Then, for $Y \sim \text{Poi}(\lambda)$ and for given k , $0 \leq k < \infty$

$$P(X_n = k) \rightarrow P(Y = k)$$

as $n \rightarrow \infty$

- ▶ Moreover, let $X_n = B_1 + B_2 + \cdots + B_n$ where B_i are independent Bernoulli RVs with parameters $p_i = p_{i,n}$.
- ▶ Let $Y_n \sim \text{Poi}(\lambda)$ where $\lambda = \lambda_n = \sum_{i=1}^n p_i$
- ▶ Then,

$$\sum_{k=0}^{\infty} |P(X_n = k) - P(Y_n = k)| \leq 2 \frac{1 - e^{-\lambda}}{\lambda} \sum_{i=1}^n p_i^2$$

Example

- ▶ A weekly lottery is conducted in which three numbers out of 25 are selected at random
- ▶ To win, you must hold exactly those three numbers. Suppose you play for n weeks where n is large
- ▶ What is the probability of winning the lottery at least once?
At least twice?

Example

- ▶ The number of weeks the player wins is $X \sim \text{Bin}(n, p)$ where $p = \frac{1}{\binom{25}{3}} = 0.00043$

- ▶ Can approximate $X \overset{\text{approx}}{\sim} \text{Poi}(\lambda)$ with $\lambda = np = 0.00043n$



$$P(X \geq 1) = 1 - P(X = 0) \approx 1 - e^{-0.00043n}$$

and

$$\begin{aligned} P(X \geq 2) &= 1 - P(X = 0) - P(X = 1) \approx \\ &1 - e^{-0.00043n} - 0.00043ne^{-0.00043n} \\ &= 1 - (1 + 0.00043n)e^{-0.00043n} \end{aligned}$$

- ▶ E.g. if $n = 5$ years (of 52 weeks each), $P(X \geq 1) = 0.106$ and $P(X \geq 2) = 0.006$

Benford's Law

- ▶ For numbers selected at random from a statistical dataset, the leading digit tends to be 1 with probability higher than 11.1% expected if the distribution was just discrete uniform on $\{1, 2, \dots, 9\}$.
- ▶ This distribution gives reasonable approximation to the first digit in randomly picked home addresses, leading digits in geological variables, baseball statistics etc.
- ▶ The fits are even better when data from apparently unrelated sources are combined.
- ▶ It is a distribution on $\{1, 2, \dots, 9\}$ with the pmf

$$p(x) = \frac{\log(x+1) - \log x}{\log 10}$$

- ▶ It is a curious fact that the mean is 3.44

Distribution of Sums and Differences

- ▶ The general question: suppose $X_1, \dots, X_n$ are n independent random variables with known distributions.
- ▶ What is the distributions of the sum $X_1 + \dots + X_n$?
- ▶ For some distributions, if all X_i have the same distribution, then the sum will also have the same distribution
- ▶ This is called **loyalty to types**.

Basic result

- ▶ Let $X_1, \dots, X_k \sim \text{Bin}(n_i, p)$, $i = 1, \dots, k$ and independent. Then,

$$\sum_{i=1}^k X_i \sim \text{Bin} \left(\sum_{i=1}^k n_i, p \right)$$

- ▶ If $X_1, \dots, X_n$ are k independent negative binomial RVs with $X_i \sim \text{NB}(r_i, p)$, then

$$\sum_{i=1}^k X_i \sim \text{NB} \left(\sum_{i=1}^k r_i, p \right)$$

- ▶ If $X_1, \dots, X_n$ are k independent Poisson random variables with $X_i \sim \text{Poi}(\lambda_i)$, then

$$\sum_{i=1}^k X_i \sim \text{Poi} \left(\sum_{i=1}^k \lambda_i \right)$$

Example

- ▶ Let $X \sim \text{Poi}(1)$, $Y \sim \text{Poi}(5)$ and $Z \sim \text{Poi}(10)$.
- ▶ X , Y , and Z are independent
- ▶ By the previous theorem $(X + Y + Z) \sim \text{Poi}(16)$
- ▶

$$\begin{aligned} P(X + Y + Z \geq 20) &= 1 - P(X + Y + Z \leq 19) \\ &= 1 - \sum_{x=0}^{19} \frac{e^{-16} 16^x}{x!} = 0.1878 \end{aligned}$$